```
a = [10, 40, 7, 50, 90]
```

Access the element 50.

```
a[3]
```

```
50
```

Access the element 90.

```
a[4]
```

```
90
```

Access the elements from 40 to 50.

```
a[1:4]                  #[start:end+1]
```

```
[40, 7, 50]
```

Access the elements from 10 to 7.

```
a[0:3]
```

```
[10, 40, 7]
```

```
b = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

Access the even elements.

```
b[1:10:2]
```

```
[2, 4, 6, 8, 10]
```

Access the odd elements.

```
b[0:10:2]
```

```
[1, 3, 5, 7, 9]
```

Print the multiples of three.

```
b[2:10:3]
```

```
[3, 6, 9]
```

Print the multiples of four.

```
b[3:10:4]
```

```
[4, 8]
```

Built-in Functions. (sum, len, max, min, sorted, round, abs)

```
c = [10, 20, 40, 7, 90]

sum(c)

167

len(c)

5

max(c)

90

min(c)

7

sorted(c)                              #by default Ascending order

[7, 10, 20, 40, 90]

m = 9.9
round(m)

10

n = -85
abs(n)

85
```

Hello, [your name]! I'm learning Python in [year].

```
name = input("enter the name")
year = input("enter the year")
print(f"Hello, {name}! I'm learning Python in {year}.")

enter the name Madhusruti
enter the year 2026

Hello, Madhusruti! I'm learning Python in 2026.
```

Here are your test scores: [45, 78,90,66]. Find total marks, highest, lowest and average.

```
test_scores = [45, 78, 90, 66]
print(f"Total marks is {sum(test_scores)}")
```

```
print(f"Highest marks is {max(test_scores)}")
print(f"Lowest marks is {min(test_scores)}")
print(f"Average marks is {sum(test_scores)/len(test_scores)}")
```

```
Total marks is 279
Highest marks is 90
Lowest marks is 45
Average marks is 69.75
```

If a student scores above 90, you say 'Excellent'.

Else if above 60, you say 'Good'.

Else say 'Try again!'

```
score = int(input("Enter the marks"))
if (score>90):
    print("Excellent")
elif (score>60):
    print("Good")
else:
    print("Try again!")
```

```
Enter the marks 95

Excellent
```

```
score = int(input("Enter the marks"))
if (score>90):
    print("Excellent")
elif (score>60):
    print("Good")
else:
    print("Try again!")
```

```
Enter the marks 65

Good
```

```
score = int(input("Enter the marks"))
if (score>90):
    print("Excellent")
elif (score>60):
    print("Good")
else:
    print("Try again!")
```

```
Enter the marks 55

Try again!
```

Score > 90 --> "Excellent"

Score >= 75 --> "Good"

Score >= 50 --> "Improvement"

Else Fail

```
Score = int(input("enter the marks"))
if(Score>90):
    print("Excellent")
elif(Score>=75):
    print("Good")
elif(Score>=50):
    print("Improvement")
else:
    print("Fail")
```

```
enter the marks 45
```

```
Fail
```

If temperature > 30 --> Hot

If temperature > 20 --> Nice weather

Else --> Cold

```
temperature = int(input("Enter the temperature"))
if (temperature>30):
    print("Hot")
elif (temperature>20):
    print("Nice weather")
else:
    print("Cold")
```

```
Enter the temperature 35
```

```
Hot
```

```
temperature = int(input("Enter the temperature"))
if (temperature>30):
    print("Hot")
elif (temperature>20):
    print("Nice weather")
else:
    print("Cold")
```

```
Enter the temperature 26
```

```
Nice weather
```

```python
temperature = int(input("Enter the temperature"))
if (temperature>30):
    print("Hot")
elif (temperature>20):
    print("Nice weather")
else:
    print("Cold")
```

```
Enter the temperature 18
```

```
Cold
```

If your bill crosses Rs.500/-, you will get a discount of 10%.

```python
bill = int(input("Enter the bill amount"))      #Case - 01
if (bill>500):
    bill = bill - 0.1*bill
    print(bill)
else:
    print(bill)
```

```
Enter the bill amount 900
```

```
810.0
```

```python
bill = int(input("Enter the bill amount"))       #Case - 02
if (bill>500):
    bill = bill - 0.1*bill
    print(bill)
else:
    print(bill)
```

```
Enter the bill amount 450
```

```
450
```

Here "bill" is a string. So when we do print("bill"), the output is bill. And when we do print(bill), the output is numerical. In this case, we treat bill as integer data type.

```python
bill = int(input("Enter the bill amount"))     #Case - 03
if (bill>500):
    bill = bill - 0.1*bill
    print("bill")
else:
    print(bill)
```

```
Enter the bill amount 900
```

```
bill
```

```
bill = int(input("Enter the bill amount"))      #Case - 04
if (bill>500):
    bill = bill - 0.1*bill
    print(bill)
else:
    print("bill")
```

```
Enter the bill amount 450

bill
```

For Loop

```
for i in range(5):                           #for loop runs from 0 to 4
  print("I Love Enhypen")
```

```
I Love Enhypen
I Love Enhypen
I Love Enhypen
I Love Enhypen
I Love Enhypen
```

```
for i in range(1,6):                      #here in range 6 (end) is --> 6-1 = 5
    print("I Love BTS")
```

```
I Love BTS
I Love BTS
I Love BTS
I Love BTS
I Love BTS
```

Print the numbers from 1 to 10.

```
for i in range(1,11):                        #here in range 11 (end) is --> 11-1 
= 10
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

Print the numbers from 6 to 11.

```
for i in range(6,12):                          #here in range 12 (end) is 
--> 12-1 = 11
    print(i)
```

```
6
7
8
9
10
11
```